សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ

Hotel Management System

បង្រៀនដោយសាស្ត្រាចារ្យ៖ Dan Vournpheng

ថ្នាក់ A4

សមាជិក

១. រ៉ុ ឌីរ៉ៈ **២. ប៉ាត មូលសក្តា** **៣. នឹម រត្ត**នៈ **៤. ទូ និចាន់** **៥. ឈឿន គុណា**

CONTENTS

Table of Contents

Objective	03
Form 1 — Login Form	04
Form 2 — Main Form	05
Form 3 — Guest Mgmt	06
Form 4 — Room Mgmt	07
Form 5 — Booking List	08
Form 6 — New Booking	09
Form 7 — Payment	10
Form 8 — Receipt	11
Form 9 — Invoice Mgmt	12
Form 10 — Housekeeping	13
Form 11 — Staff Mgmt	14
Form 12 — Daily Report	15
Design Patterns (7 patterns)	16 – 23
Conclusion	24
Demo	25

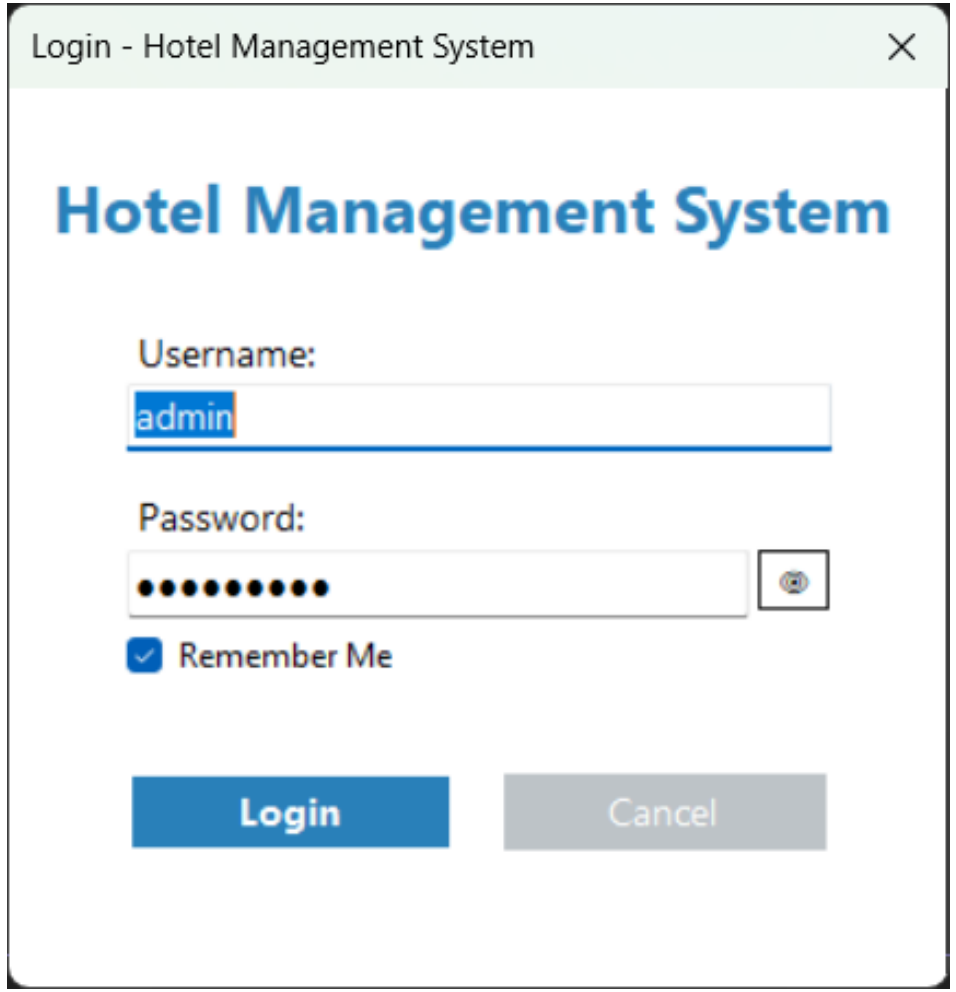
Objective

- Desktop hotel management platform built in C# Windows Forms
- Covers complete lifecycle: booking → check-in → checkout → payment
- Role-based access: Admin · Receptionist · Housekeeping
- 12 operational forms · Layered architecture · 7 design patterns

FORM 1

Login Form

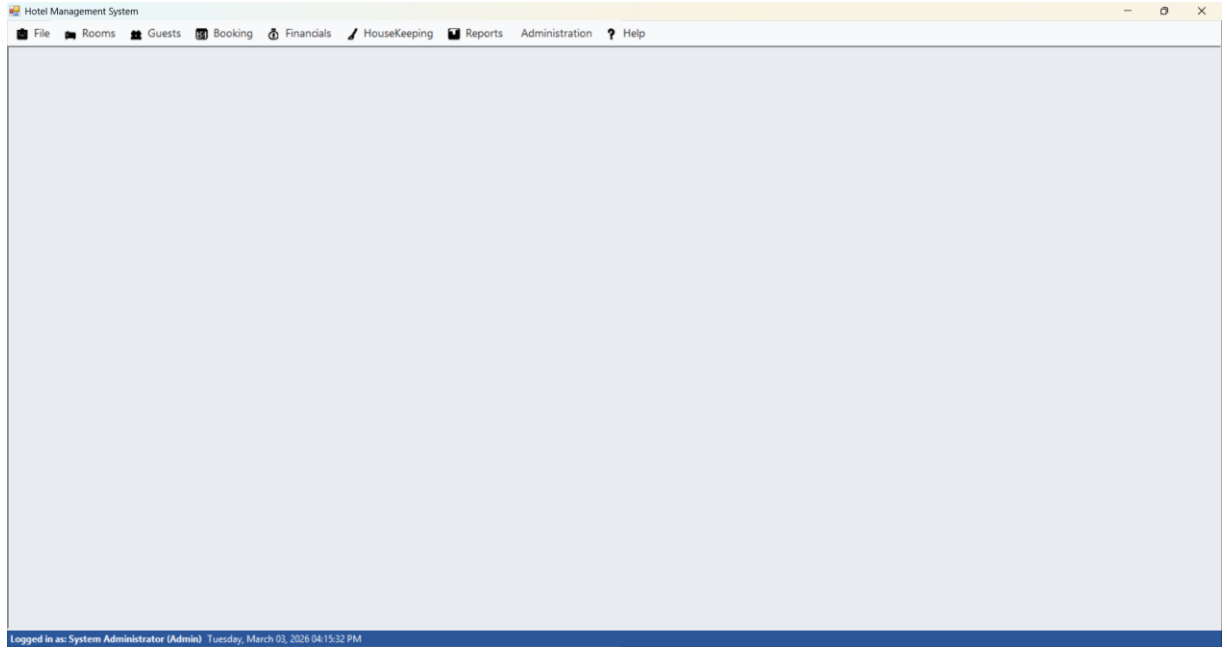
- Secure login — SHA-256 password hashing with random salt
- Three roles: Admin / Receptionist / Housekeeping
- "Remember Me" — credentials persisted in app settings
- Wrong credentials → error shown, password field cleared



FORM 2

Main Form — MDI Hub

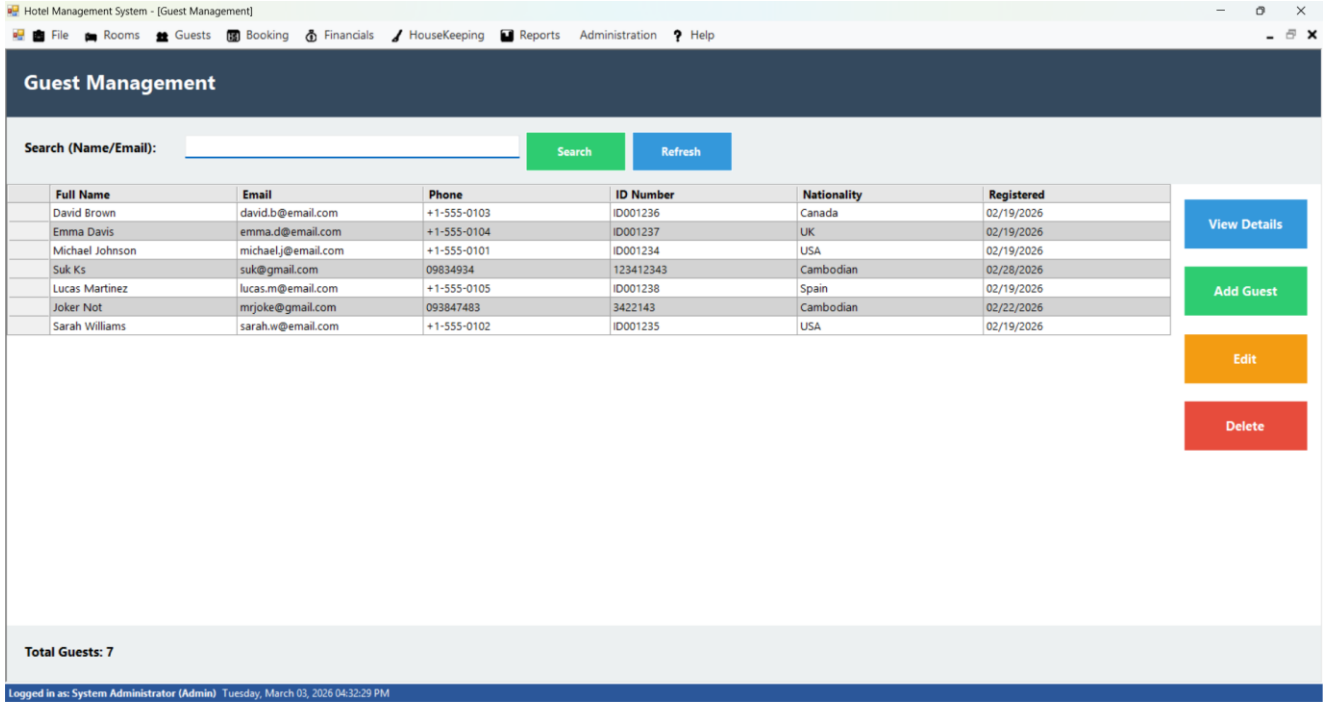
- Central navigation hub for all modules
- Menu bar: Booking · Guest · Room · Housekeeping · Reports
- Status bar shows logged-in user, role, and live clock
- Observer pattern initialized here on startup



FORM 3

Guest Management

- Add, edit, and soft-delete guest records
- Search by name, email, or phone number
- View full guest profile and booking history

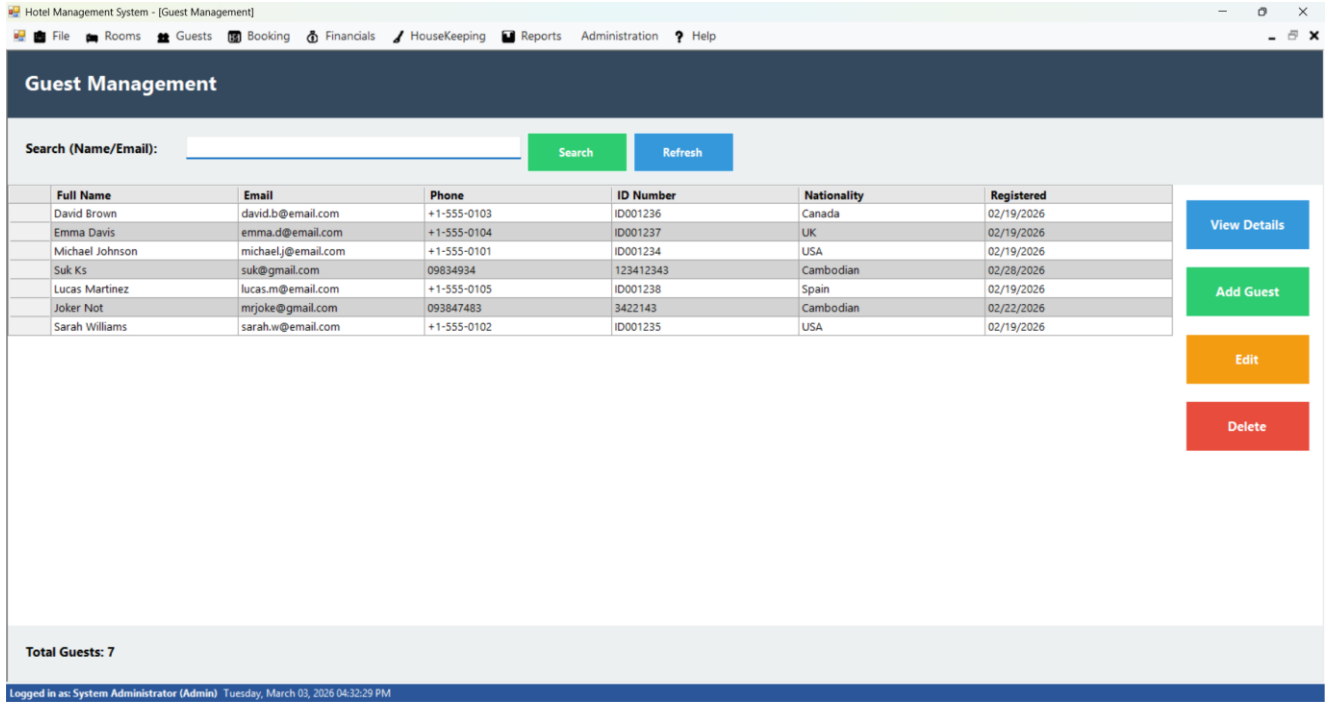


Full Name	Email	Phone	ID Number	Nationality	Registered
David Brown	david.b@email.com	+1-555-0103	ID001236	Canada	02/19/2026
Emma Davis	emma.d@email.com	+1-555-0104	ID001237	UK	02/19/2026
Michael Johnson	michael.j@email.com	+1-555-0101	ID001234	USA	02/19/2026
Suk Ks	suk@gmail.com	09834934	123412343	Cambodian	02/28/2026
Lucas Martinez	lucas.m@email.com	+1-555-0105	ID001238	Spain	02/19/2026
Joker Not	mrjoke@gmail.com	093847483	3422143	Cambodian	02/22/2026
Sarah Williams	sarah.w@email.com	+1-555-0102	ID001235	USA	02/19/2026

FORM 4

Room Management

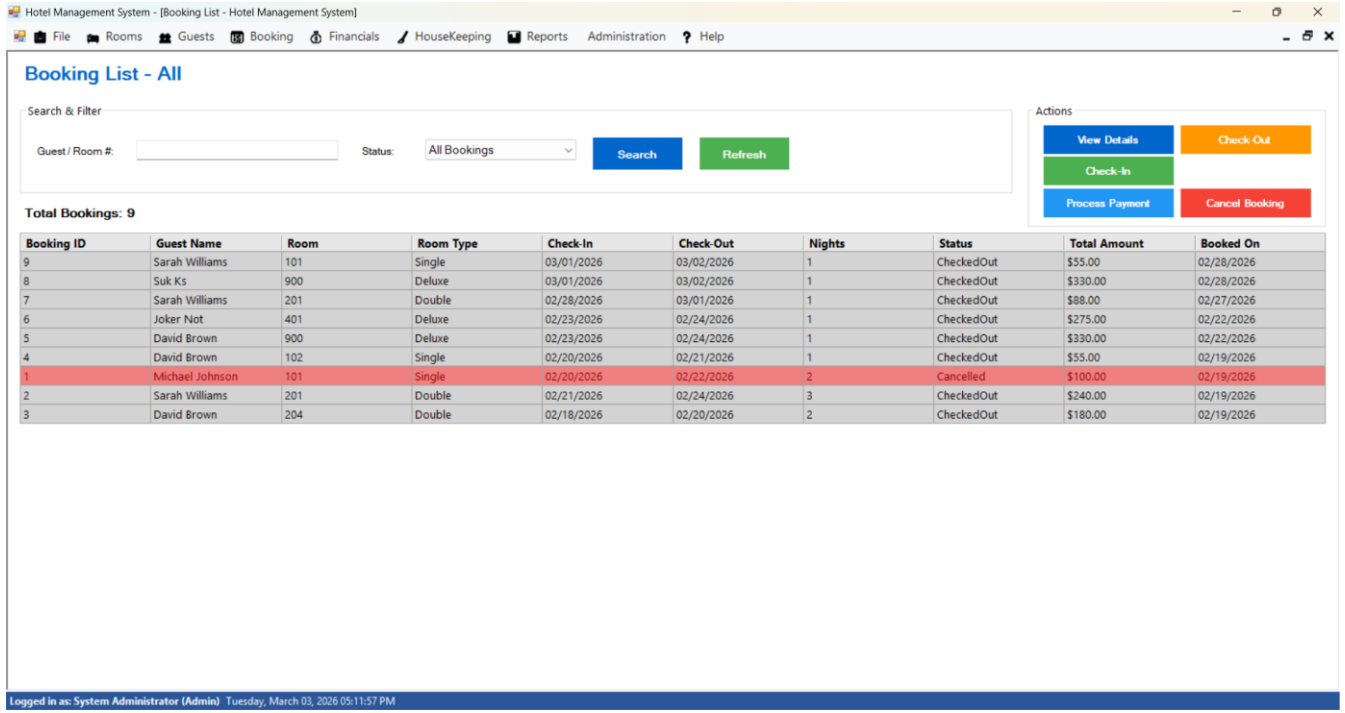
- Four room types: Single · Double · Suite · Deluxe
- Track status: Available · Occupied · Reserved · Maintenance
- Color-coded visual status dashboard
- Room objects created via Factory Pattern



FORM 5

Booking List

- All bookings shown with color-coded status rows
- Actions: Check-In · Check-Out · Process Payment · Cancel
- Check-Out auto-generates invoice + triggers housekeeping task

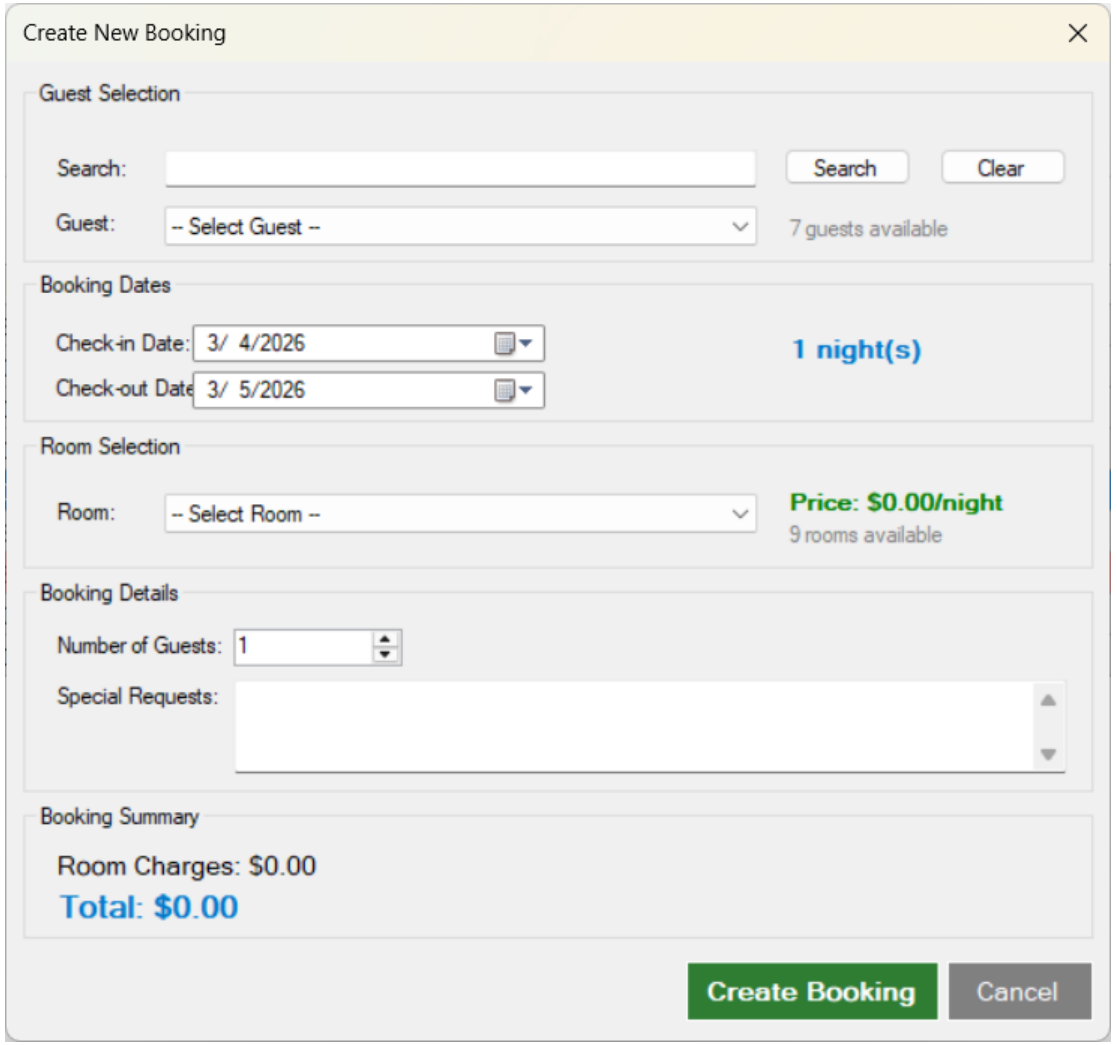


Booking ID	Guest Name	Room	Room Type	Check-In	Check-Out	Nights	Status	Total Amount	Booked On
9	Sarah Williams	101	Single	03/01/2026	03/02/2026	1	CheckedOut	$55.00	02/28/2026
8	Suk Ks	900	Deluxe	03/01/2026	03/02/2026	1	CheckedOut	$330.00	02/28/2026
7	Sarah Williams	201	Double	02/28/2026	03/01/2026	1	CheckedOut	$88.00	02/27/2026
6	Joker Not	401	Deluxe	02/23/2026	02/24/2026	1	CheckedOut	$275.00	02/22/2026
5	David Brown	900	Deluxe	02/23/2026	02/24/2026	1	CheckedOut	$330.00	02/22/2026
4	David Brown	102	Single	02/20/2026	02/21/2026	1	CheckedOut	$55.00	02/19/2026
1	Michael Johnson	101	Single	02/20/2026	02/22/2026	2	Cancelled	$100.00	02/19/2026
2	Sarah Williams	201	Double	02/21/2026	02/24/2026	3	CheckedOut	$240.00	02/19/2026
3	David Brown	204	Double	02/18/2026	02/20/2026	2	CheckedOut	$180.00	02/19/2026

FORM 6

New Booking

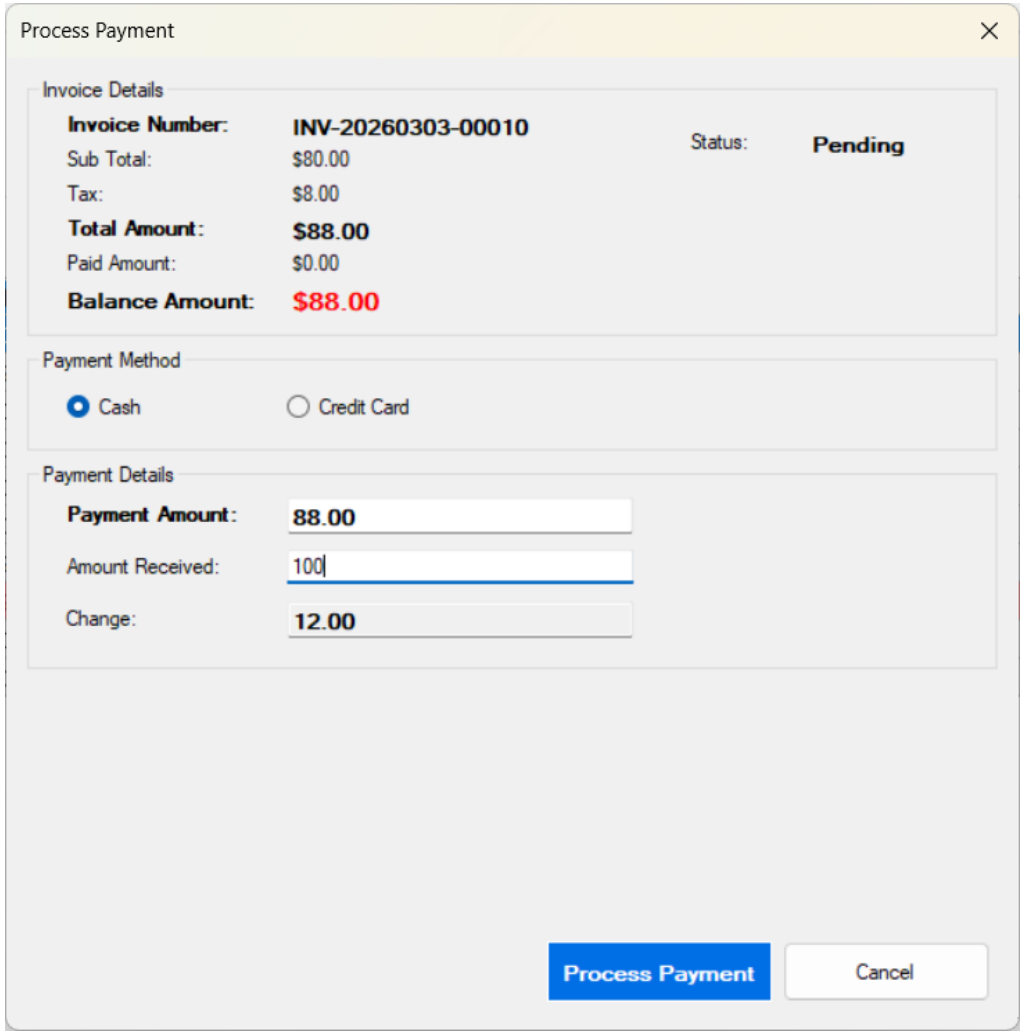
- Select guest, room, and date range
- Auto-calculates: Base Price × Nights + 10% tax
- Add-ons: Breakfast · Extra Bed · Airport Transfer (Decorator)
- Entire booking workflow managed via Facade Pattern



FORM 7

Payment Form

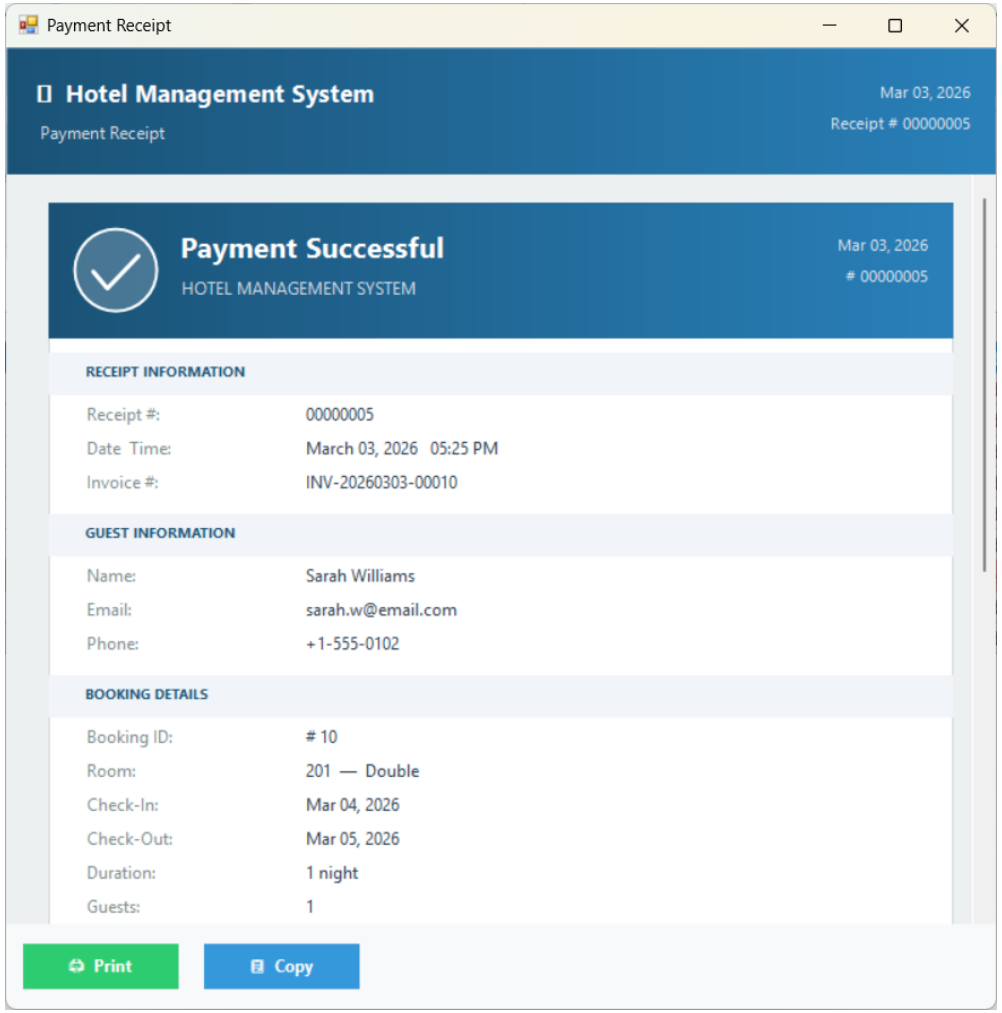
- Cash — enter amount, change calculated automatically
- Credit Card — Luhn validation + simulated gateway (95% success)
- Strategy Pattern: payment method switched at runtime
- Success → Receipt Form displayed automatically



FORM 8

Receipt Form

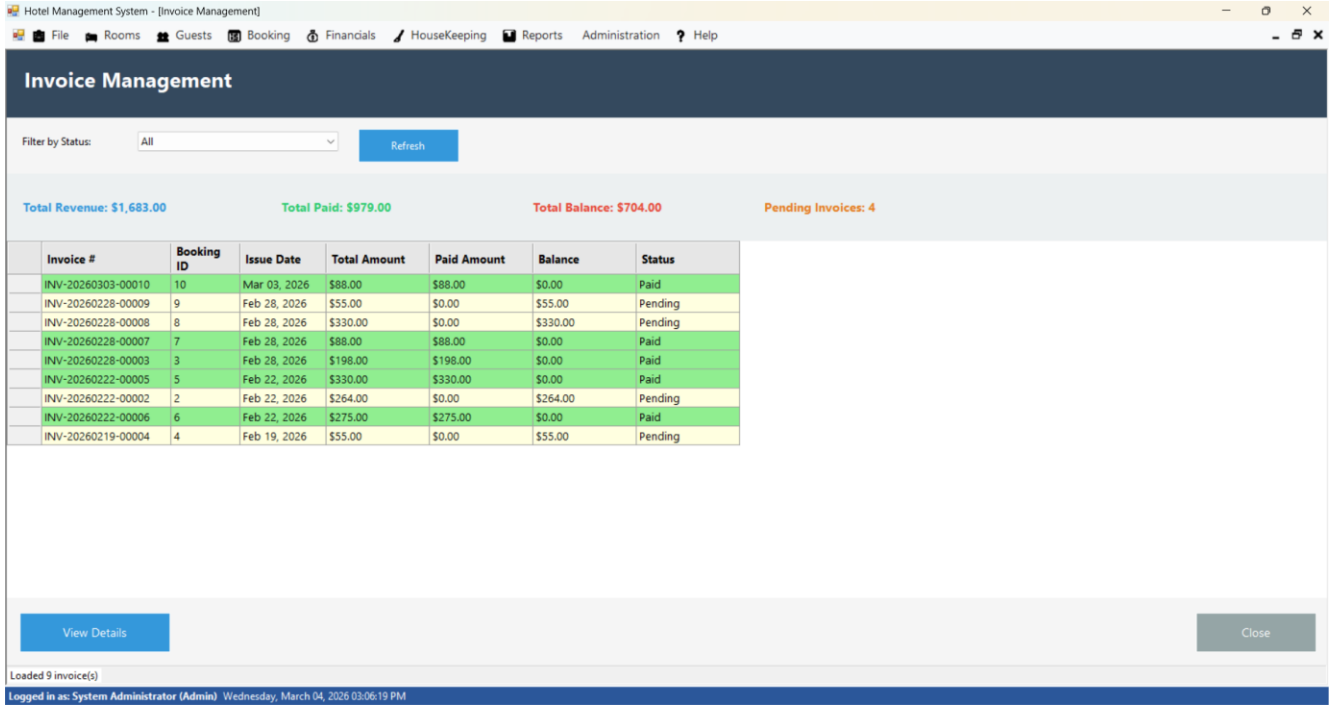
- Styled receipt shown after successful payment
- Sections: Guest info · Booking details · Financial breakdown
- Status badge: Paid / Partially Paid / Pending
- Actions: Print · Export PDF · Copy to Clipboard



FORM 9

Invoice Management

- Invoices auto-generated on guest checkout
- Subtotal + 10% tax + discounts applied automatically
- Status tracking: Pending · Paid · Partially Paid · Cancelled

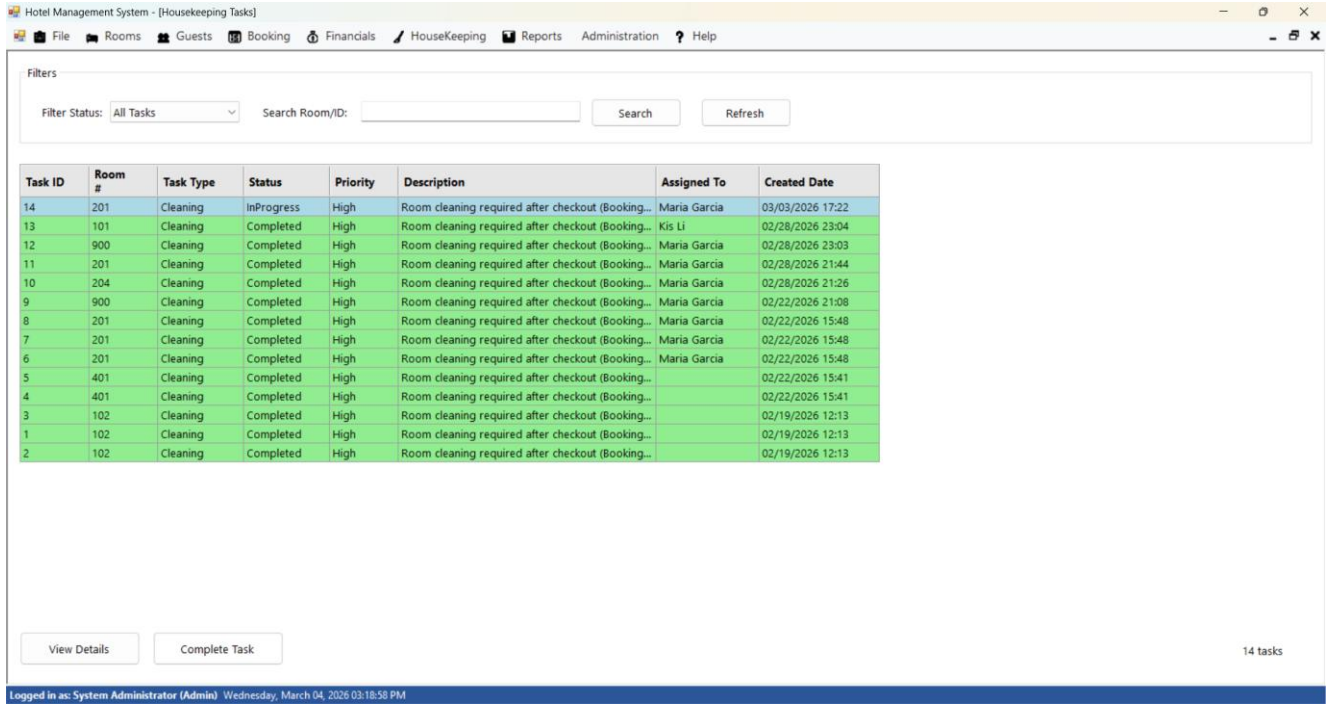


Invoice #	Booking ID	Issue Date	Total Amount	Paid Amount	Balance	Status
INV-20260303-00010	10	Mar 03, 2026	$88.00	$88.00	$0.00	Paid
INV-20260228-00009	9	Feb 28, 2026	$55.00	$0.00	$55.00	Pending
INV-20260228-00008	8	Feb 28, 2026	$330.00	$0.00	$330.00	Pending
INV-20260228-00007	7	Feb 28, 2026	$88.00	$88.00	$0.00	Paid
INV-20260228-00003	3	Feb 28, 2026	$198.00	$198.00	$0.00	Paid
INV-20260222-00005	5	Feb 22, 2026	$330.00	$330.00	$0.00	Paid
INV-20260222-00002	2	Feb 22, 2026	$264.00	$0.00	$264.00	Pending
INV-20260222-00006	6	Feb 22, 2026	$275.00	$275.00	$0.00	Paid
INV-20260219-00004	4	Feb 19, 2026	$55.00	$0.00	$55.00	Pending

FORM 10

Housekeeping Tasks

- Cleaning tasks auto-created on checkout (Observer Pattern)
- Smart assignment — least-busy staff selected automatically
- Status workflow: Pending → In Progress → Completed

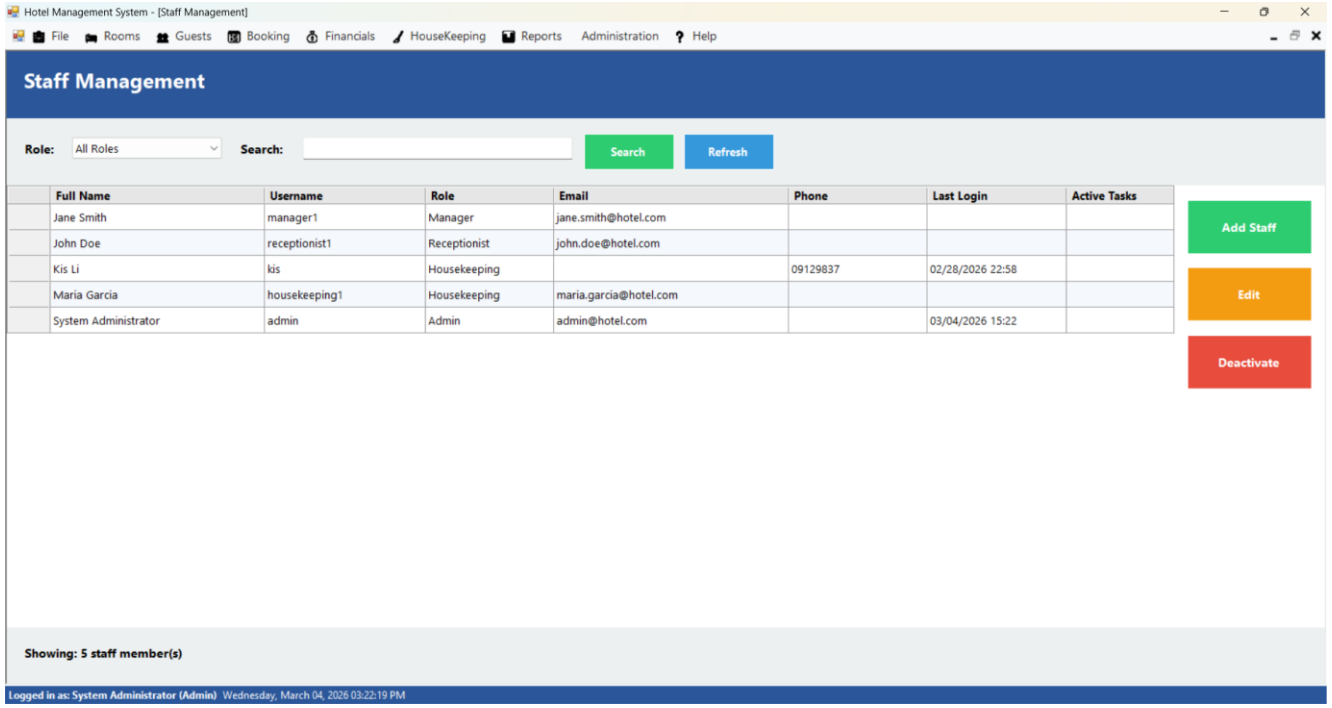


Task ID	Room #	Task Type	Status	Priority	Description	Assigned To	Created Date
14	201	Cleaning	InProgress	High	Room cleaning required after checkout (Booking...	Maria Garcia	03/03/2026 17:22
13	101	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Kis Li	02/28/2026 23:04
12	900	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/28/2026 23:03
11	201	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/28/2026 21:44
10	204	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/28/2026 21:26
9	900	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/22/2026 21:08
8	201	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/22/2026 15:48
7	201	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/22/2026 15:48
6	201	Cleaning	Completed	High	Room cleaning required after checkout (Booking...	Maria Garcia	02/22/2026 15:48
5	401	Cleaning	Completed	High	Room cleaning required after checkout (Booking...		02/22/2026 15:41
4	401	Cleaning	Completed	High	Room cleaning required after checkout (Booking...		02/22/2026 15:41
3	102	Cleaning	Completed	High	Room cleaning required after checkout (Booking...		02/19/2026 12:13
1	102	Cleaning	Completed	High	Room cleaning required after checkout (Booking...		02/19/2026 12:13
2	102	Cleaning	Completed	High	Room cleaning required after checkout (Booking...		02/19/2026 12:13

FORM 11

Staff Management

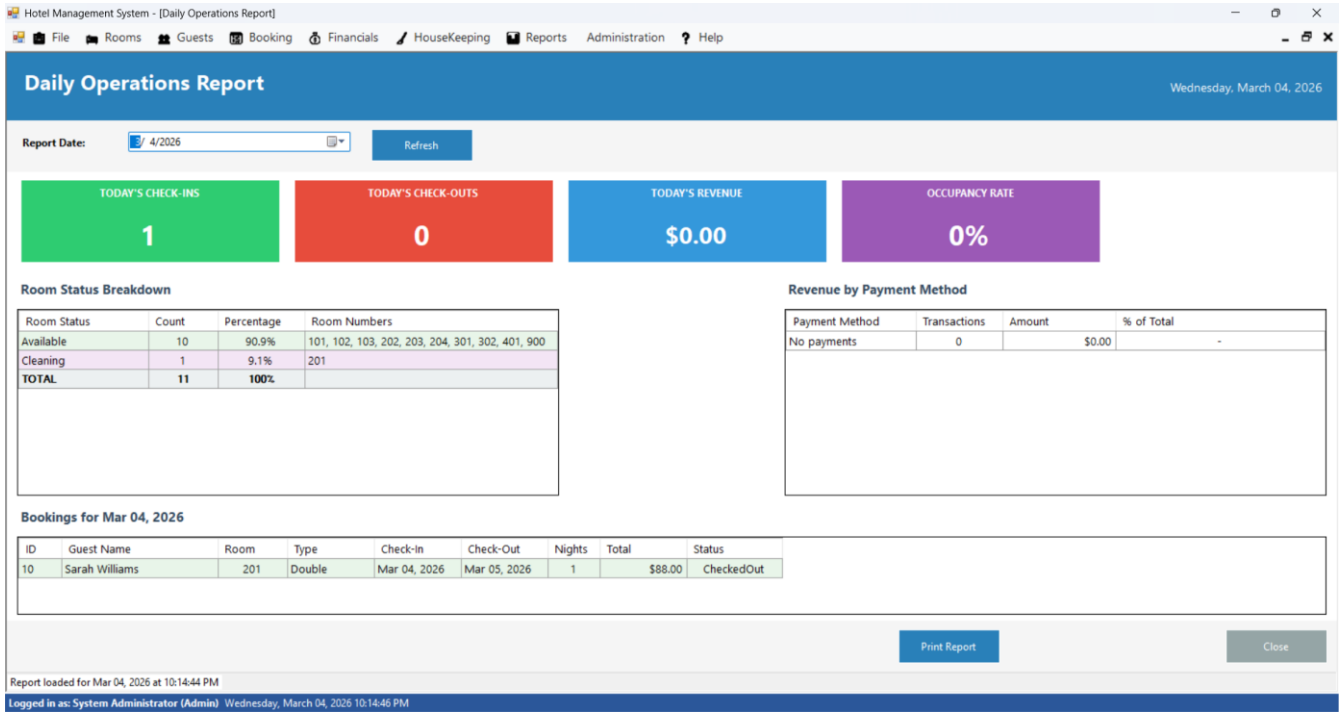
- CRUD operations for all staff accounts
- Role assignment: Admin · Receptionist · Housekeeping
- Secure password management with SHA-256 hashing



FORM 12

Daily Operations Report

- Dashboard of daily hotel metrics
- Tracks occupancy, bookings, payments, and outstanding tasks



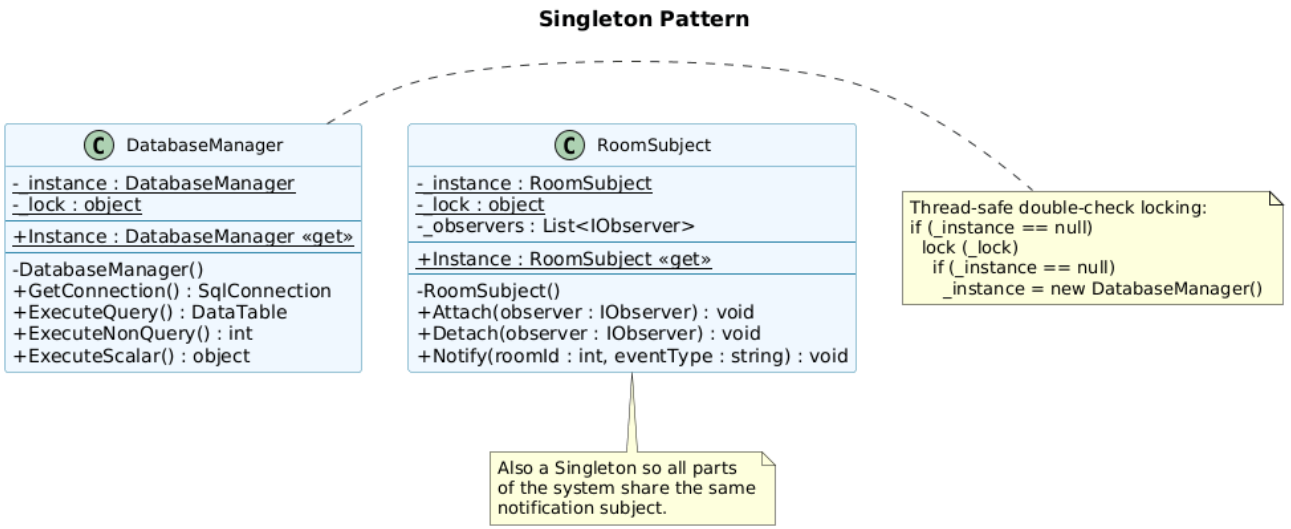
Design Patterns — 7 Implemented

- 1. Singleton — single shared instance
- 2. Factory — creates correct room type objects
- 3. Strategy — interchangeable payment algorithms
- 4. Observer — auto-triggers housekeeping on checkout
- 5. Repository — clean, centralised data access layer
- 6. Facade — simplified booking workflow
- 7. Decorator — stackable room add-on services

PATTERN 1

Singleton

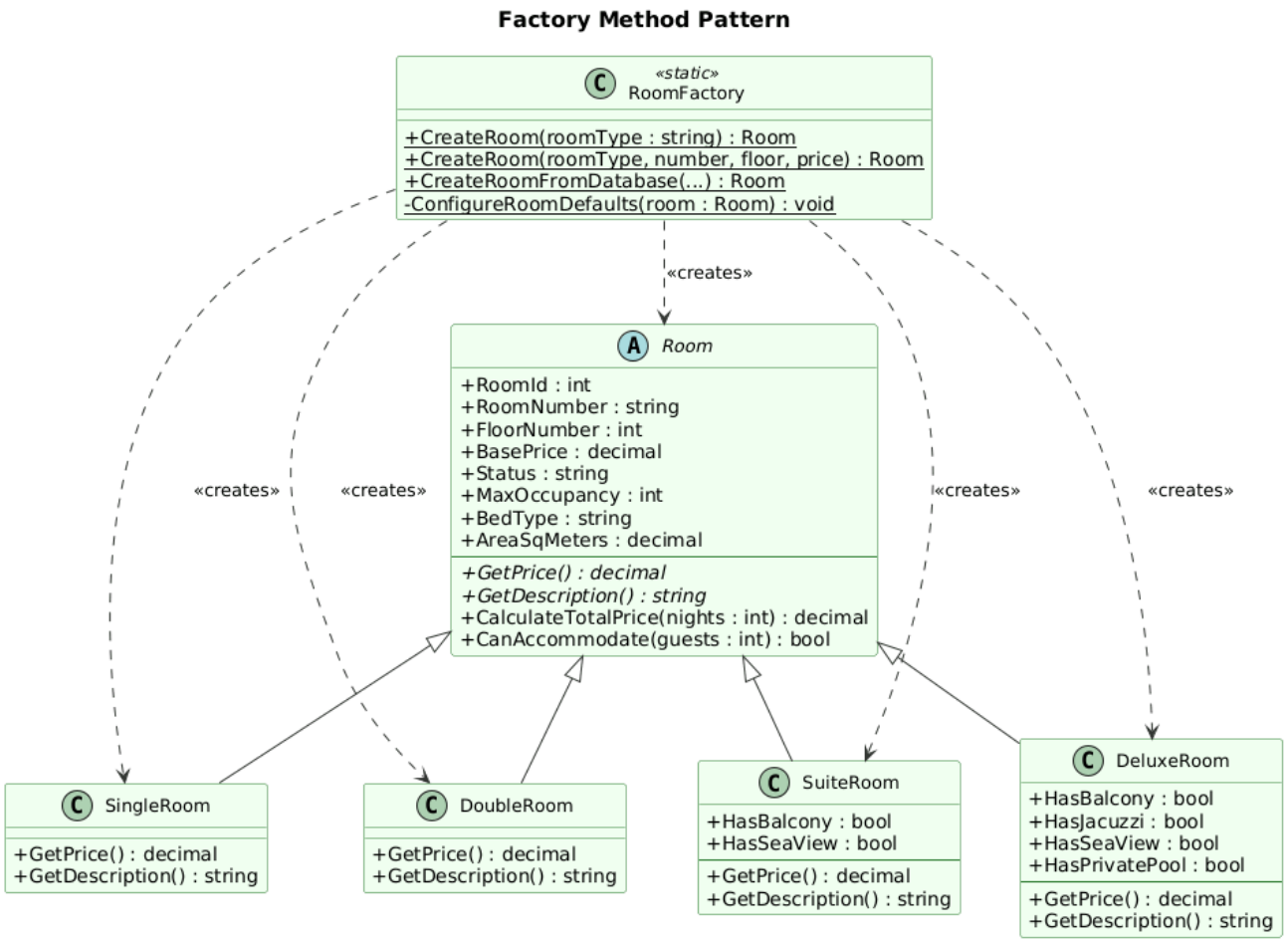
- Guarantees only one instance exists throughout the app
- DatabaseManager — single thread-safe connection pool
- RoomSubject — one central observable event hub
- Double-check locking ensures thread safety



PATTERN 2

Factory Method

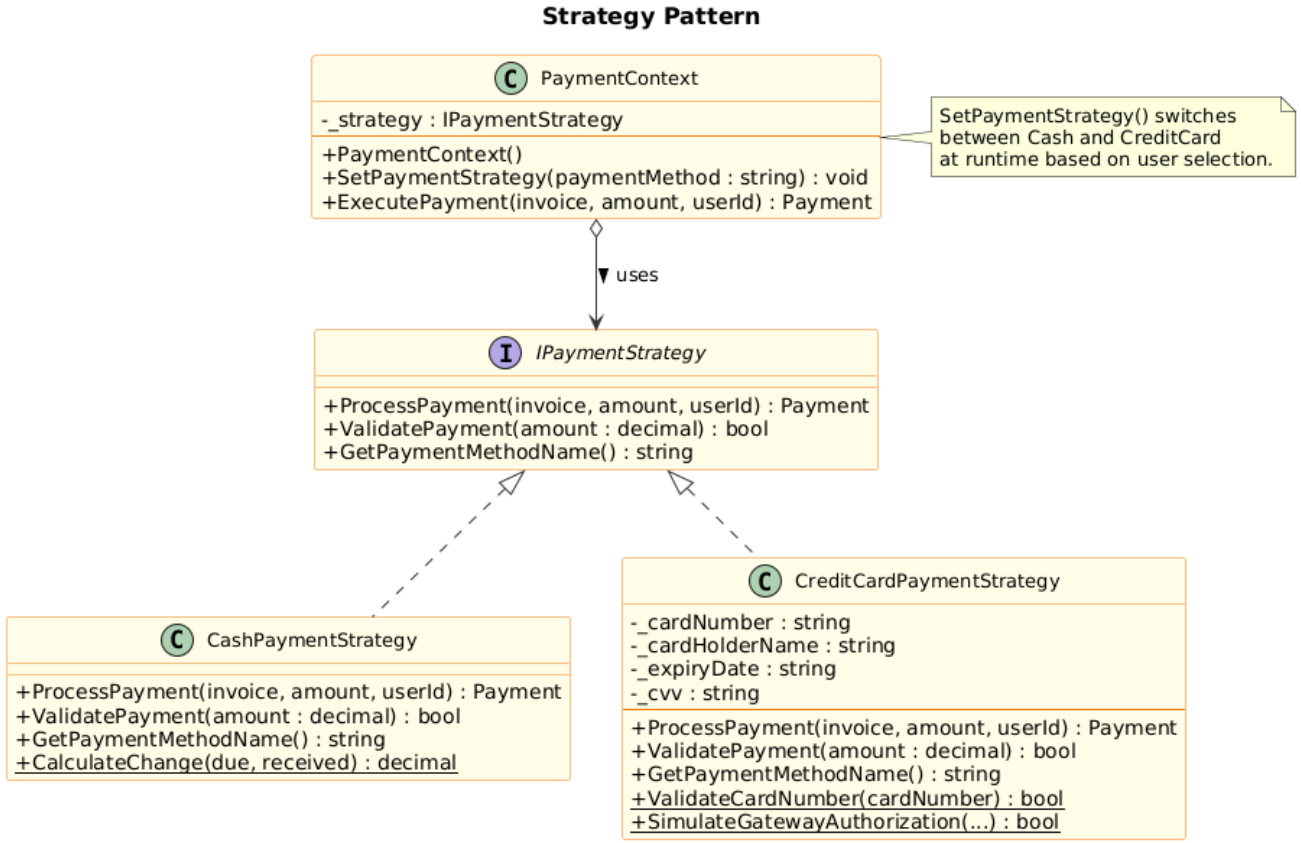
- RoomFactory.CreateRoom(type) returns the correct subclass
- 4 types: Single · Double · Suite · Deluxe
- Each type carries preset defaults: price, bed, area, occupancy
- UI never references concrete room classes directly



PATTERN 3

Strategy

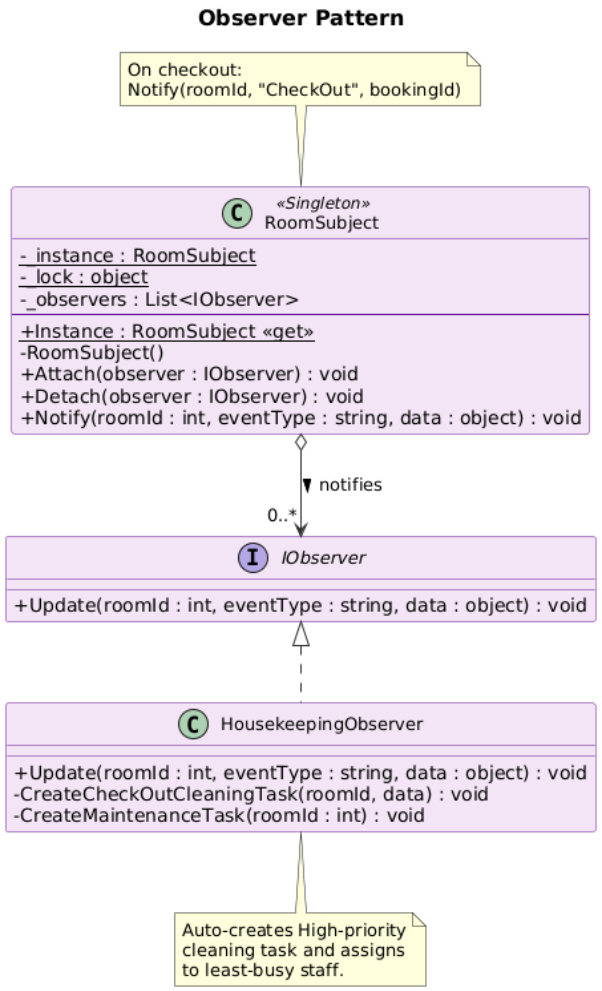
- Defines interchangeable payment processing algorithms
- CashPaymentStrategy — records transaction and computes change
- CreditCardPaymentStrategy — Luhn check + gateway authorisation
- PaymentContext switches strategy at runtime based on user choice



PATTERN 4

Observer

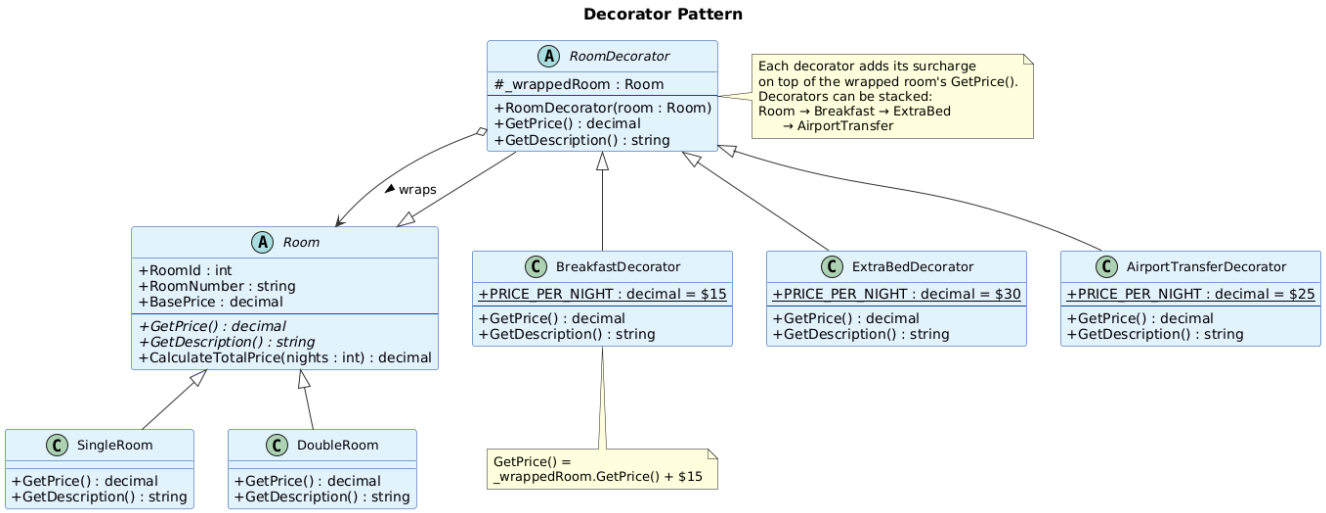
- RoomSubject broadcasts room events to all registered observers
- On checkout → HousekeepingObserver auto-creates cleaning task
- Smart assignment: least-busy housekeeping staff selected
- Checkout code stays decoupled — only calls Notify()



PATTERN 5

Decorator

- Dynamically adds services to any room type at booking time
- Breakfast +$15/night · Extra Bed +$30/night · Airport Transfer +$25/night
- Decorators stack: Double($80) + Breakfast + Extra Bed = $125/night
- Open/Closed — new services added without touching existing code



Conclusion

- Covers complete hotel lifecycle end-to-end
- 7 design patterns applied to real architectural challenges
- Secure: SHA-256 hashing · Role-based access · Luhn validation
- Extensible: new payment methods and services add without touching existing code

Project Demo 🎬